

Basics of Electrical Engineering (EE113ATD)

Unit-IV

Transformers (single phase): principle of operation, Types and construction, EMF equation, losses, efficiency and regulation expressions.

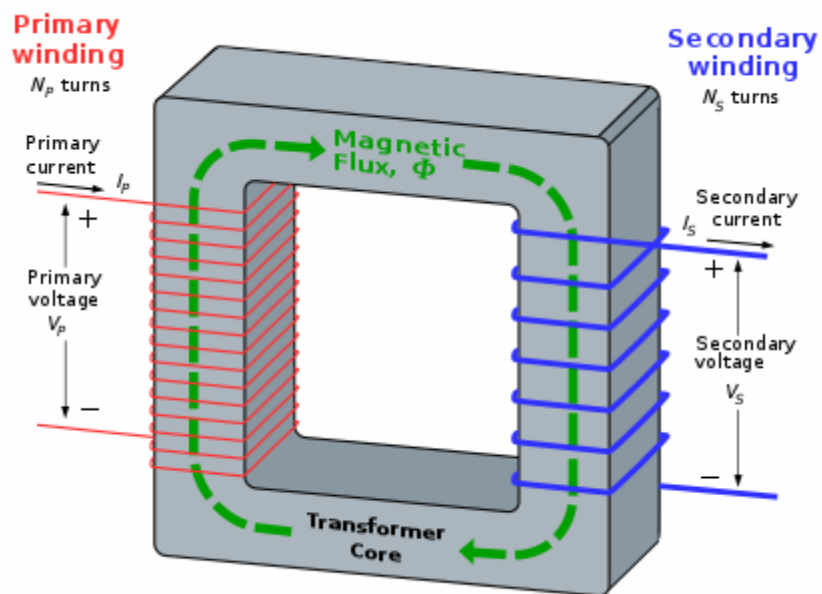
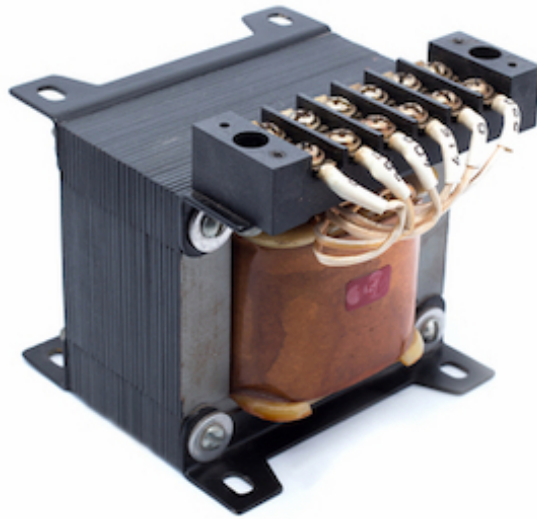
Three-phase induction Motors: Concept of rotating magnetic field, Principle, constructional features of squirrel cage and wound rotor, slip and torque characteristics.

Alternators: Types, principle of operation.

Objectives & Outcomes

1. **Understand and explain** the principle, construction, types, and EMF equation of single-phase transformers, and **analyze** their performance in terms of losses, efficiency, and regulation.
2. **Illustrate and apply** the concept of rotating magnetic field, and **differentiate** the constructional features and operational aspects of squirrel cage and wound rotor three-phase induction motors.
3. **Interpret and analyze** slip–torque characteristics to evaluate the performance of three-phase induction motors for practical applications.
4. **Classify and explain** the types and principle of operation of alternators, and **relate** their significance in electrical power generation and systems.

Single Phase Transformers



Operating Principles and Construction What is a Transformer?

A **transformer** is a static piece of equipment used either for raising or lowering the voltage of an AC supply with a corresponding decrease or increase in current.

The use of transformers in transmission system is shown in the Figure below.

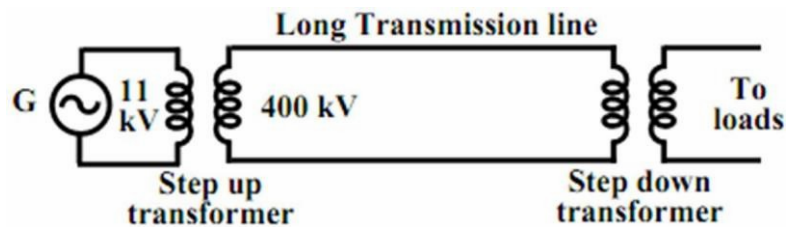


Fig.3-1 A simple single phase power system.

Principle of Operation

A transformer in its simplest form will consist of a rectangular laminated magnetic structure on which two coils of different number of turns are wound as shown in Figure 3.2a.

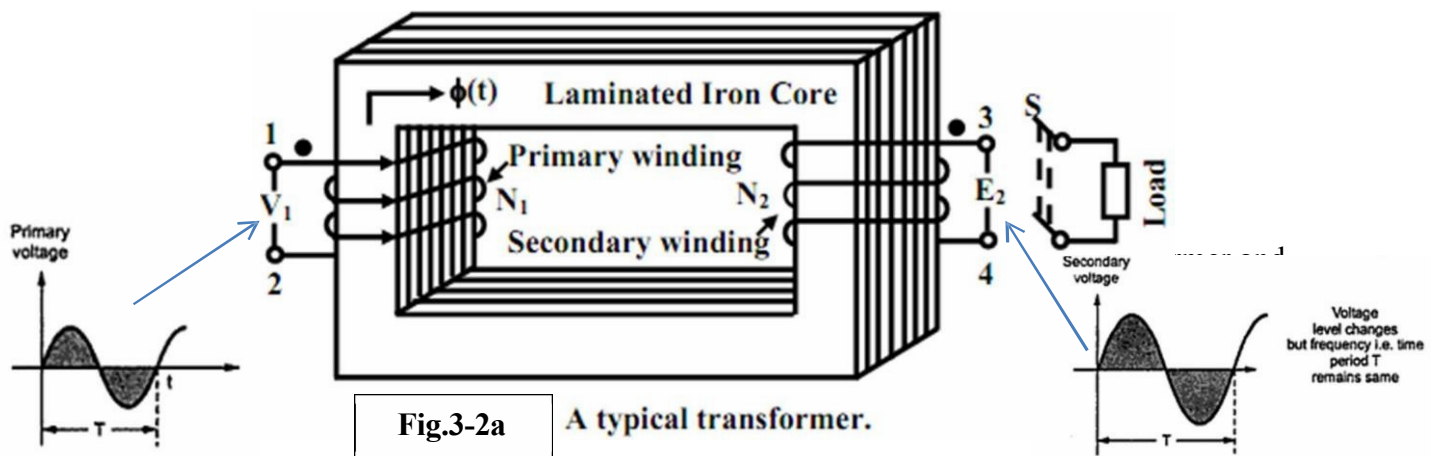


Fig.3-2a A typical transformer.

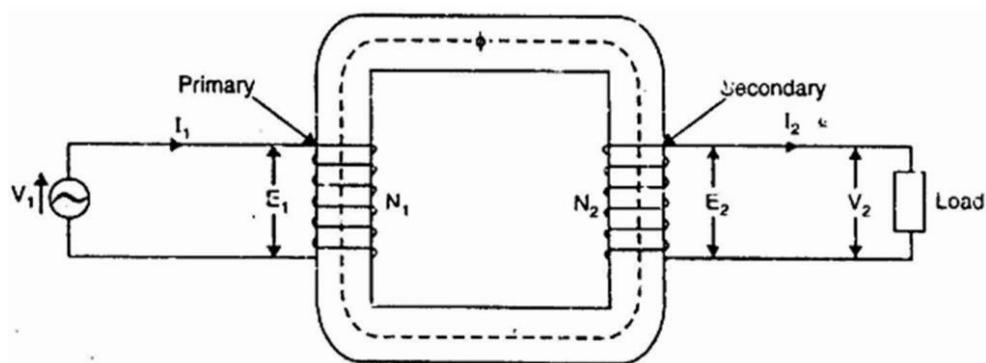


Fig.3-2b

Depending upon the number of turns of the primary (N_1) and secondary (N_2), an alternating emf (E_2) is induced in the secondary. This induced emf (E_2) in the secondary causes a secondary current I_2 . Consequently, terminal voltage V_2 will appear across the load. If $V_2 > V_1$, it is called a **step up-transformer**. On the other hand, if $V_2 < V_1$, it is called a **step-down transformer**.

When an alternating voltage V_1 is applied to the primary, an alternating flux Φ is set up in the core. This alternating flux links both the windings and induces emfs E_1 and E_2 in them according to **Faraday's laws of electromagnetic induction**. The emf E_1 is termed as primary emf and emf E_2 is termed as Secondary emf.

Clearly,
$$E_1 = -N_1 \frac{d\phi}{dt}$$

and
$$E_2 = -N_2 \frac{d\phi}{dt}$$

$$\therefore \frac{E_2}{E_1} = \frac{N_2}{N_1}$$

Note that magnitudes of E_2 and E_1 depend upon the number of turns on the secondary and primary respectively. If $N_2 > N_1$, then $E_2 > E_1$ (or $V_2 > V_1$) and we get a step-up transformer. On the other hand, if $N_2 < N_1$, then $E_2 < E_1$ (or $V_2 < V_1$) and we get a step-down transformer. If load is connected across the secondary winding, the secondary e.m.f. E_2 will cause a current I_2 to flow through the load. Thus, a transformer enables us to transfer a.c. power from one circuit to another with a change in voltage level.

The following points may be noted carefully:

- (i) The transformer action is based on the laws of *electromagnetic induction*.
- (ii) There is no electrical connection between the primary and secondary.
- (iii) There is no change in frequency i.e., output power has the same frequency as the input power.

Can DC Supply be used for Transformers?

The DC supply cannot be used for the transformers. This is because the transformer works on the principle of *mutual induction*, for which current in one coil must change uniformly. If DC supply is given, the current will not change due to constant supply and transformer will not work

There can be saturation of the core due to which transformer draws very large current from the supply when connected to DC.

Thus DC supply should not be connected to the transformers.

Construction

We usually design a power transformer so that it approaches the characteristics of an ideal transformer. To achieve this, following design features are incorporated:

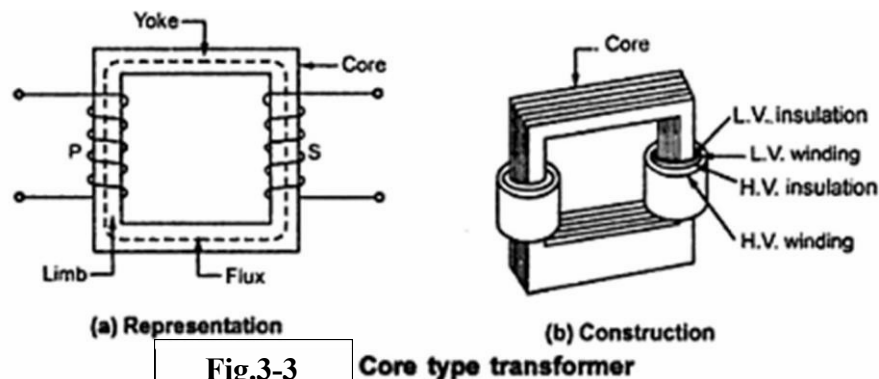
- (i) The core is made of silicon steel which has low hysteresis loss and high permeability. Further, core is laminated in order to reduce eddy current loss. These features considerably reduce the iron losses and the no-load current.
- (ii) Instead of placing primary on one limb and secondary on the other, it is a usual practice to wind one-half of each winding on one limb. This ensures tight coupling between the two windings. Consequently, leakage flux is considerably reduced.

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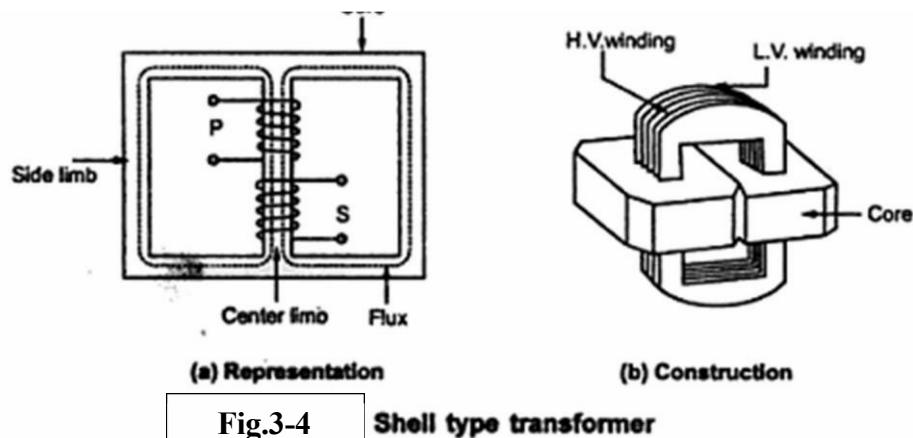
- (iii) The winding resistances are minimized to reduce Copper loss and resulting rise in temperature and to ensure high efficiency.

Transformers are of two types: (i) core-type transformer (see Fig.3-3) and (ii) shell-type transformer (see Fig.3-4).

Core-Type Transformer: In a core-type transformer, half of the primary winding and half of the secondary winding are placed round each limb to reduce the leakage flux.



Shell-Type Transformer: This method of construction involves the use of a double magnetic circuit. Both the windings are placed round the central limb to ensure a low-reluctance flux path.



Comparison of Core and Shell Type Transforms

Core Type	Shell Type
The winding encircles the core.	The core encircles most part of the winding
It has single magnetic circuit	It has double magnetic circuit
The core has two limbs	The core has three limbs
The cylindrical coils are used.	The multilayer disc or sandwich type coils are used.
The winding are uniformly distributed on two limbs hence natural cooling is effective	The natural cooling does not exist as the windings are surrounded by the core.
Preferred for low voltage transformers.	Preferred for high voltage transformers.

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EMF Equation of a Transformer

Consider that an alternating voltage V_1 of frequency f is applied to the primary as shown in Fig. 3-2b. The sinusoidal flux Φ produced by the primary can be represented as:

$$\phi = \phi_m \sin \omega t$$

The instantaneous e.m.f. e_1 induced in the primary is

$$\begin{aligned} e_1 &= -N_1 \frac{d\phi}{dt} = -N_1 \frac{d}{dt} (\phi_m \sin \omega t) \\ &= -\omega N_1 \phi_m \cos \omega t = -2\pi f N_1 \phi_m \cos \omega t \\ &= 2\pi f N_1 \phi_m \sin(\omega t - 90^\circ) \end{aligned} \quad (i)$$

It is clear from the above equation that maximum value of induced e.m.f. in the primary is

$$E_{m1} = 2\pi f N_1 \phi_m$$

The r.m.s. value E_1 of the primary e.m.f. is

$$E_1 = \frac{E_{m1}}{\sqrt{2}} = \frac{2\pi f N_1 \phi_m}{\sqrt{2}}$$

or
$$E_1 = 4.44 f N_1 \phi_m$$

Similarly
$$E_2 = 4.44 f N_2 \phi_m$$

In an ideal transformer, $E_1 = V_1$ and $E_2 = V_2$.

Note. It is clear from exp. (i) above that e.m.f. E_1 induced in the primary lags behind the flux ϕ by 90° . Likewise, e.m.f. E_2 induced in the secondary lags behind flux ϕ by 90° .

Voltage Transformation Ratio (K)

From the above equations of induced emf, we have,

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

The constant K is called voltage transformation ratio. Thus if $K = 5$ (i.e. $N_2/N_1 = 5$), then

$$E_2 = 5 E_1.$$

Concept of Ideal Transformer

A transformer is said to be ideal if it satisfies following properties:

- i) It has no losses.
- ii) Its windings have zero resistance.
- iii) Leakage flux is zero i.e. 100 % flux produced by primary links with the secondary.
- iv) Permeability of core is so high that negligible current is required to establish the flux in it.

NOTE:

For an ideal transformer, the primary applied voltage V_1 is same as the primary induced emf E_1 as there are no voltage drops.

For ideal transformer:

- (i) $E_1 = V_1$ and $E_2 = V_2$ as there is no voltage drop in the windings.

$$\therefore \frac{E_2}{E_1} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

- (ii) there are no losses. Therefore, volt-amperes input to the primary are equal to the output volt-amperes i.e.

$$V_1 I_1 = V_2 I_2$$

$$\text{or} \quad \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

Hence, currents are in the inverse ratio of voltage transformation ratio. This simply means that if we raise the voltage, there is a corresponding decrease of current.

Volt-Ampere Rating

Transformer rating is specified as the product of voltage and current and called VA rating.

$$\text{kVA rating of a transformer} = \frac{V_1 I_1}{1000} = \frac{V_2 I_2}{1000}$$

The full load primary and secondary currents which indicate the safe maximum values of currents which transformer windings can carry can be given as:

$I_1 \text{ full load} = \frac{\text{kVA rating} \times 1000}{V_1} \quad \dots \text{ (1000 to convert kVA to VA)}$
$I_2 \text{ full load} = \frac{\text{kVA rating} \times 1000}{V_2}$

Ideal Transformer on No Load

Consider an ideal transformer in Fig. 3-5. For no load $I_2 = 0$. I_1 is just necessary to produce flux in the core, which is called **magnetizing** current denoted as I_m . I_m is very small and lags V_1 by 90° as the winding is purely inductive.

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According to Lenz's law, the induced e.m.f opposes the cause producing it which is supply voltage V_1 . Hence E_1 and E_2 are in antiphase with V_1 but equal in magnitude and E_1 and E_2 are in phase.

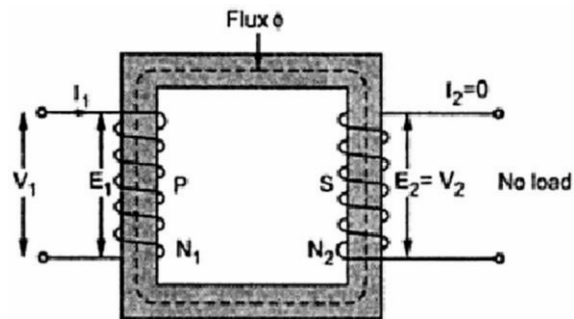
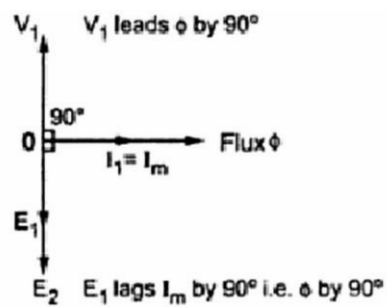


Fig.3-5

This can be illustrated in the phase diagram as shown below:



Phasor diagram for ideal transformer on no load

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Losses in a Transformer

The power losses in a transformer are of two types, namely;

1. **Core or Iron losses**
2. **Copper losses**

NOTE: The above losses appear in the form of heat and produce (i) *an increase in temperature* and (ii) a *drop in efficiency*.

A- Core or Iron losses (P_i)

These consist of *hysteresis and eddy current losses* and occur in the transformer core due to the alternating flux. These can be determined by open-circuit test (see next sections).

$$\text{Hysteresis loss, } = k_h f B_m^{1.6} \text{ watts/m}^3$$

$$\text{Eddy current loss, } = k_e f^2 B_m^2 t^2 \text{ watts/m}^3$$

Both hysteresis and eddy current losses depend upon

Maximum flux density B_m in the core and

Supply frequency f .

NOTE: Since transformers are connected to constant-frequency, constant voltage supply, both f and B_m are constant. Hence, core or iron losses are practically the same at all loads. Hence,

$$\begin{aligned} \text{Iron or Core losses, } P_i &= \text{Hysteresis loss} + \text{Eddy current loss} \\ &= \text{Constant losses} \end{aligned}$$

NOTE: The hysteresis loss can be minimized *by using steel of high silicon content* whereas eddy current loss can be reduced *by using core of thin laminations*.

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B- Copper losses (P_C)

These losses occur in both the primary and secondary windings due to their ohmic resistance.
These can be determined by short-circuit test

Total copper Cu losses:

$$\begin{aligned} P_C &= I_1^2 R_1 + I_2^2 R_2 \\ &= I_1^2 R_{01} \text{ or } I_2^2 R_{02} \end{aligned}$$

Hence, total losses in a transformer are:

$$\begin{aligned} \text{Total losses in a transformer} &= P_1 + P_C \\ &= \text{Constant losses} + \text{Variable losses} \end{aligned}$$

Efficiency of a Transformer

Like any other electrical machine, the efficiency of a transformer is defined as
the ratio of output power (in watts or kW) to input power (watts or kW) i.e.

$$\text{Efficiency} = \frac{\text{Output power}}{\text{Input power}}$$

In practice, open-circuit and short-circuit tests are carried out to find the efficiency,

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}} = \frac{\text{Output}}{\text{Output} + \text{Losses}}$$

NOTE: The losses can be determined by transformer tests.

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Condition for Maximum Efficiency

$$\text{Output power} = V_2 I_2 \cos \phi_2$$

If R_{02} is the total resistance of the transformer referred to secondary, then,

$$\text{Total Cu loss, } P_C = I_2^2 R_{02}$$

$$\text{Total losses} = P_i + P_C$$

$$\begin{aligned} \text{Transformer } \eta &= \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_{02}} \\ &= \frac{V_2 \cos \phi_2}{V_2 \cos \phi_2 + P_i / I_2 + I_2 R_{02}} \end{aligned}$$

For a load of given pf, efficiency depends upon load current I_2 . Hence, the efficiency to be maximum *the denominator should be minimum* i.e.

$$\frac{d}{dI_2} (V_2 \cos \phi_2 + P_i / I_2 + I_2 R_{02}) = 0$$

$$\text{or} \quad 0 - \frac{P_i}{I_2^2} + R_{02} = 0$$

$$\text{or} \quad P_i = I_2^2 R_{02}$$

i.e., Iron losses = Copper losses

Hence efficiency of a transformer will be maximum when copper losses are equal to constant or iron losses.

From above, the load current I_2 corresponding to maximum efficiency is:

$$I_2 = \sqrt{\frac{P_i}{R_{02}}}$$

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NOTE: In a transformer, iron losses are constant whereas copper losses are variable. In order to obtain **maximum efficiency**, the load current should be such that total Cu losses become equal to iron losses.

Output kVA corresponding to Maximum Efficiency Let P_C = Copper

losses at full-load kVA

P_i = Iron losses

x = Fraction of full-load kVA at which efficiency is maximum Total Cu

losses = $x^2 \times P_C$

for maximum efficiency

$$x^2 \times P_C = P_i$$

or

$$x = \sqrt{\frac{P_i}{P_C}} = \sqrt{\frac{\text{Iron loss}}{\text{F.L. Cu loss}}}$$

Output kVA corresponding to maximum efficiency:

$$\text{Full - load kVA} \times \sqrt{\frac{\text{Iron loss}}{\text{F.L. Cu loss}}}$$

NOTE: The value of kVA at which the efficiency is maximum, is independent of pf of the load.

Efficiency from Transformer Tests

The full-load efficiency of the transformer at any pf can be obtained as:

$$\text{F.L. efficiency, } \eta_{\text{F.L.}} = \frac{\text{Full - load VA} \times \text{p.f.}}{(\text{Full - load VA} \times \text{p.f.}) + P_i + P_C}$$

where:

P_i = Iron loss can be obtained from open-circuit test

P_C = Copper loss can be obtained from short-circuit test

F.L. = Full Load

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Also the efficiency for any load,

$$\begin{aligned}\text{Corresponding total losses} &= P_i + x^2 P_C \\ \text{Corresponding } \eta_x &= \frac{(\text{xx Full - load VA}) \times \text{p.f.}}{(\text{xx Full - load VA} \times \text{p.f.}) + P_i + x^2 P_C}\end{aligned}$$

where xx= Fraction of full-load

NOTE: Iron loss remains the same at all loads.

Voltage Regulation of a Transformer:

Definition: The voltage regulation is defined as the change in the magnitude of receiving and sending voltage of the transformer. The voltage regulation determines the ability of the transformer to provide the constant voltage for variable loads.

When the transformer is loaded with continuous supply voltage, the terminal voltage of the transformer varies. The variation of voltage depends on the load and its power factor.

Mathematically, the voltage regulation is represented as:

$$\text{Voltage Regulation} = \frac{E_2 - V_2}{E_2}$$

$$\% \text{ Voltage Regulation} = \frac{E_2 - V_2}{E_2} \times 100$$

where,

E_2 – secondary terminal voltage at no load V_2 –
secondary terminal voltage at full load

The voltage regulation by considering the primary terminal voltage of the transformer is expressed as,

$$\% \text{ Voltage Regulation} = \frac{V_1 - E_1}{V_1} \times 100$$

Problems on Transformers

Q1	A single phase, 20 kVA transformer has 1000 primary turns and 2500 secondary turns. The net cross sectional area of the core is 100 cm^2. When the primary winding is connected to 500V, 50 Hz supply, calculate (i) the maximum value of the flux density in the core (ii) the voltage induced in the secondary winding and (iii) the primary and secondary full load currents. i) $E_1 = 4.44 f \phi_m N_1$ i.e. $500 = 4.44 \times 50 \times \phi_m \times 1000$ $\therefore \phi_m = 2.252 \times 10^{-3} \text{ Wb}$ $B_m = \frac{\phi_m}{a} = \frac{2.252 \times 10^{-3}}{100 \times 10^{-4}} = 0.2252 \text{ Wb/m}^2$ ii) $\frac{E_2}{E_1} = \frac{N_2}{N_1}$ i.e. $\frac{E_2}{500} = \frac{2500}{1000}$, $\therefore E_2 = 1250 \text{ V}$ iii) $I_1 = \frac{\text{kVA} \times 1000}{E_1} = \frac{20 \times 1000}{500} = 40 \text{ A}$, $I_2 = \frac{20 \times 1000}{1250} = 16 \text{ A}$
Q2	Find the number of turns on the primary and secondary side of a 440/230 V, 50 Hz, single phase transformer, if the net area of a cross section of the core is 30 cm^2 and the maximum value of the flux density is 1 Wb/m^2. $E_1 = 4.44 f \phi_m N_1 = 4.44 f B_m a N_1$ i.e., $440 = 4.44 \times 50 \times 1 \times 30 \times 10^{-4} \times N_1$ $\therefore N_1 = 660.67 \text{ turns}$ $\frac{E_2}{E_1} = \frac{N_2}{N_1}$ i.e. $\frac{230}{440} = \frac{N_2}{660.67}$, $\therefore N_2 = 345.35 \text{ turns}$

Q3	The primary winding of a 25 kVA transformer has 200 turns and is connected to 230V, 50Hz supply. The secondary turns are 50. Calculate i) No load secondary induced emf ii) Full load primary and secondary currents iii) The flux density in the core, if the cross section of the core is 60 cm². i) $\frac{E_2}{E_1} = \frac{N_2}{N_1}$ i.e. $\frac{E_2}{230} = \frac{50}{200}$, $\therefore E_2 = 57.5 \text{ V}$ ii) $I_1 = \frac{25 \times 1000}{230} = 108.7 \text{ A}$ $\frac{I_1}{I_2} = \frac{N_2}{N_1}$ i.e. $\frac{108.7}{I_2} = \frac{50}{200}$, $\therefore I_2 = 434.8 \text{ A}$ iii) $E_1 = 4.44 f \phi_m N_1$ i.e. $230 = 4.44 \times 50 \times \phi_m \times 200$ $\therefore \phi_m = 5.18 \times 10^{-3} \text{ Wb}$, $B_m = \frac{\phi_m}{a} = \frac{5.18 \times 10^{-3}}{60 \times 10^{-4}} = 0.863 \text{ Wb/m}^2$
Q4	A 100 kVA, 50 Hz single phase transformer has a turns ratio of 1000/ 250. The primary winding is connected to 500V, 50 Hz supply. Find the secondary open circuit voltage and the max. value of the flux in the core. $\frac{E_2}{E_1} = \frac{N_2}{N_1}$ i.e. $\frac{E_2}{500} = \frac{250}{1000}$ $\therefore E_2 = 125 \text{ V}$ $E_1 = 4.44 f \phi_m N_1$ i.e. $500 = 4.44 \times 50 \times \phi_m \times 1000$ $\therefore \phi_m = 2.25 \times 10^{-3} \text{ Wb} = 2.25 \text{ mWb}$
Q5	A single phase, 50 Hz core type transformer has square cores of 20 cm side. The permissible maximum flux density is 1 Wb/m². Calculate the number of turns per limb on the high and low voltage sides for a 3000/ 220 V ratio. $E_1 = 4.44 f \phi_m N_1 = 4.44 f B_m a N_1$ i.e. $3000 = 4.44 \times 50 \times 1 \times (20 \times 20 \times 10^{-4}) N_1$ $\therefore N_1 = 337.82 \text{ turns} \approx 338 \text{ turns}$ $N_2 = 337.82 \times \frac{220}{3000} = 24.77 \approx 25 \text{ turns.}$ Usually, in a transformer $\frac{1}{2}$ of the primary turns and $\frac{1}{2}$ of the secondary turns are wound on each limb. $\therefore N_1 / \text{limb} = \frac{338}{2} = 169$, $N_2 / \text{limb} = \frac{25}{2} = 12.5$

Q6	The design requirements of a 6000/450 V, 50 Hz core type transformer are: approximate e.m.f./turn = 15 V, max. flux density = 1 Wb/m². Calculate suitable number of primary and secondary turns and the net cross sectional area of the core. $N_1 = \frac{6000}{15} = 400 \quad \text{and} \quad N_2 = \frac{450}{15} = 30$ $E_1 = 4.44 f B_m a N_1 \quad \text{i.e.} \quad 6000 = 4.44 \times 50 \times 1 \times a \times 400$ $\therefore a = 0.0676 \text{ m}^2 = 676 \text{ cm}^2$
Q7	A 10 kVA, 400/200 V, 50 Hz, single phase transformer has a full load copper loss of 200 W and has a full load efficiency of 96 % at 0.8 p.f. lagging. Determine the iron loss. What would be the efficiency at half of the full load and unity p.f.? Total full load losses = Input – Output $= \left(\frac{10 \times 1000 \times 0.8}{0.96} \right) - (10 \times 1000 \times 0.8)$ $= 333.33 \text{ W}$ Iron loss = 333.33 – 200 = 133.33 W Efficiency at half of the full load and u.p.f. is given by, $\eta_{1/2FL} = \frac{1/2 \times 10 \times 1000 \times 1}{1/2 \times 10 \times 1000 \times 1 + 133.33 + (1/2)^2 \times 200}$ $= \frac{5000}{5000 + 133.33 + 50}$ $= 0.9646 \quad \text{or} \quad 96.46\%$

Q8	A 600 kVA transformer has an efficiency of 92% at full load, unity p.f. and half full load, 0.9 p.f. Determine its efficiency at 75% of full load and 0.9 p.f. $0.92 = \frac{600 \times 1000 \times 1}{600 \times 1000 \times 1 + W_i + W_{cu}}$ $\therefore W_i + W_{cu} = 52,173.9 \text{ W} \quad \text{--- (1)}$ Also $0.92 = \frac{1/2 \times 600 \times 1000 \times 0.9}{1/2 \times 600 \times 1000 \times 0.9 + W_i + (1/2)^2 W_{cu}}$ $\therefore W_i + 0.25 W_{cu} = 23,478.26 \text{ W} \quad \text{--- (2)}$ Solving equations (1) and (2) we get, $W_i = 13,913.03 \text{ W}$ and $W_{cu} = 38,260.87 \text{ W}$. $\therefore \eta_{0.75f, 0.9p.f} = \frac{0.75 \times 600 \times 1000 \times 0.9}{0.75 \times 600 \times 1000 \times 0.9 + 13,913.03 + (0.75)^2 \times 38,260.87}$ $= 0.9196 \text{ or } 91.96\%$
Q9	The maximum efficiency at full load and unity p.f of a single phase 25 kVA, 500/1000V, 50Hz transformer is 98%. Determine its efficiency at (i) 75% load, 0.9 p.f. and (ii) 50% load, 0.8 p.f. Maximum efficiency occurs when $W_i = W_{cu}$ $\therefore 0.98 = \frac{1 \times 25 \times 1000 \times 1}{1 \times 25 \times 1000 \times 1 + W_i + W_{cu}}$ $\therefore W_i = W_{cu} = 255.1 \text{ W}$ $\eta_{0.75f, 0.9p.f} = \frac{0.75 \times 25 \times 1000 \times 0.9}{0.75 \times 25 \times 1000 \times 0.9 + 255.1 + (0.75)^2 \times 255.1}$ $= 0.9769 \text{ or } 97.69\%$ $\eta_{0.5f, 0.8p.f} = \frac{0.5 \times 25 \times 1000 \times 0.8}{0.5 \times 25 \times 1000 \times 0.8 + 255.1 + (0.5)^2 \times 255.1}$ $= 0.969 \text{ or } 96.9\%$

Q10

A 10 kVA, 400/200 V, single phase transformer has a maximum efficiency η 98 % at 90 % of the full load at 0.8 p.f. Find its efficiency at full load at 0.6 p.f.

Output at 90% full load and 0.8 p.f.

$$= 10 \times 0.9 \times 0.8 = 7.2 \text{ kW}$$

$$\text{Input at 90\% full load} = \frac{\text{Output}}{\eta} = \frac{7.2}{0.98} = 7.347 \text{ kW}$$

$$\begin{aligned} \text{Total losses} &= \text{Input} - \text{Output} = 7.347 - 7.2 \\ &= 0.147 \text{ kW} \end{aligned}$$

When the efficiency is maximum, $W_i = W_{cu}$

$$\therefore W_i = \frac{0.147}{2} = 0.0735 \text{ kW}$$

$$W_{cu} \text{ at 90\% full load} = 0.0735 \text{ kW}$$

$$W_{cu} \text{ at full load} = \left(\frac{1}{0.9}\right)^2 \times 0.0735 = 0.0907 \text{ kW}$$

$\therefore$ Efficiency at full load, 0.6 p.f. is given by,

$$\eta_{F.L} = \frac{10 \times 0.6}{10 \times 0.6 + 0.0735 + 0.0907} = 0.9734 \text{ or } 97.34\%$$

At full load

$$0.9 = \frac{1 \times 500}{500 + W_i + W_{cu}}, \quad \therefore W_i + W_{cu} = 55.56 \text{ watts} \quad \text{--- (1)}$$

At $\frac{1}{2}$ full load

$$0.9 = \frac{250}{250 + W_i + \left(\frac{1}{2}\right)^2 W_{cu}}$$

$$\therefore W_i + \frac{W_{cu}}{4} = 27.78 \text{ watts}$$

$$\text{i.e. } 4 W_i + W_{cu} = 111.12 \text{ watts} \quad \text{--- (2)}$$

Solving equations (1) and (2), we get

$$W_i = 18.52 \text{ watts} \quad \text{and} \quad W_{cu} = 37.04 \text{ watts}$$

$$\begin{aligned} \eta_{0.75fl} &= \frac{0.75 \times 500}{0.75 \times 500 + 18.52 + (0.75)^2 \times 37.04} \\ &= 0.9056 \quad \text{or} \quad 90.56\% \end{aligned}$$

THREE PHASE INDUCTION MOTORS

The induction machine was invented by NIKOLA TESLA in 1888. Right from its inception its ease of manufacture and its robustness have made it a very strong candidate for electromechanical energy conversion. It is available from fractional horsepower ratings to megawatt levels. It finds very wide usage in all various application areas. The induction machine is an AC electromechanical energy conversion device. The machine interfaces with the external world through two connections (ports) one mechanical and one electrical. The mechanical port is in the form of a rotating shaft and the electrical port is in the form of terminals where AC supply is connected. There are machines available to operate from three phase or single phase electrical input.

CONSTRUCTION AND TYPES OF THREE PHASE INDUCTION MOTORS

A 3 phase induction motor consists of two major parts:

- A stator
- A rotor

Stator of 3 Phase Induction Motor

The **stator** of three phase induction motor is made up of numbers of slots to construct a 3 phase winding circuit which is connected with 3 phase AC source. The three-phase winding are arranged in such a manner in the slots that they produce one rotating magnetic field when the three-phase AC supply source is connected. The stator is made up of various stampings with slots to carry three-phase windings. It is wound for a distinct number of poles. The windings are geometrically divided 120 degrees separated. The Figure below represents the stator of three phase induction motor.



Rotor of 3 Phase Induction Motor

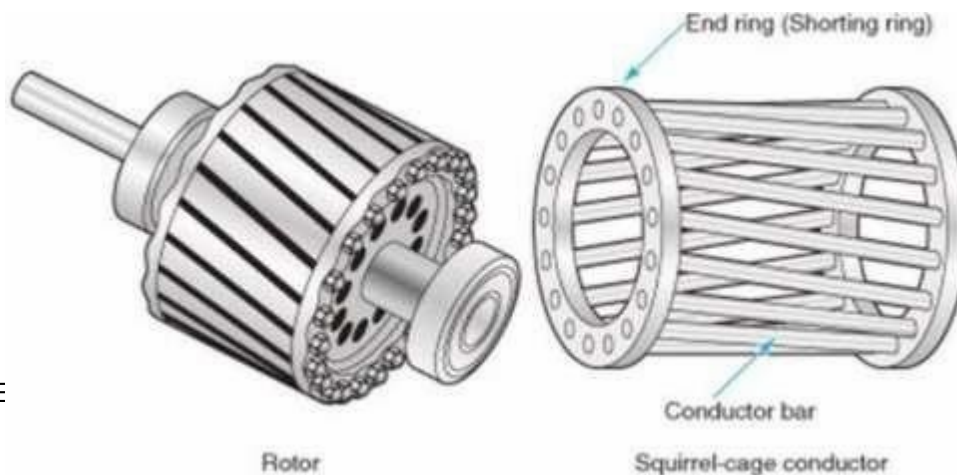
The rotor is the rotating part of the electromagnetic circuit. The rotor of three phase induction motor consists of a cylindrical laminated core with parallel slots that can carry conductors. The conductors are heavy copper or aluminum bars fitted in each slot and short-circuited by the end rings. The slots are not exactly made parallel to the axis of the shaft but are slotted a little skewed because this arrangement reduces magnetic humming noise and can avoid stalling of the motor.

Two types of rotors are used in Induction motors: **Squirrel cage rotor and Wound rotor**. No DC field current is required to run the machine. Rotor voltage is induced in the rotor windings rather than being physically connected by wires.

An induction motor has the same physical stator as a synchronous machine with an alternate rotor development. Induction motor might be worked as either motors or generator. On the other hand, they are fundamentally used as induction motors.

Squirrel cage motors are widely used due to their rugged construction and simple design.. In this motor, the rotor includes a cylindrical core that can be laminated & includes some slots on the external periphery. These slots are not comparable & they are twisted by some angles.

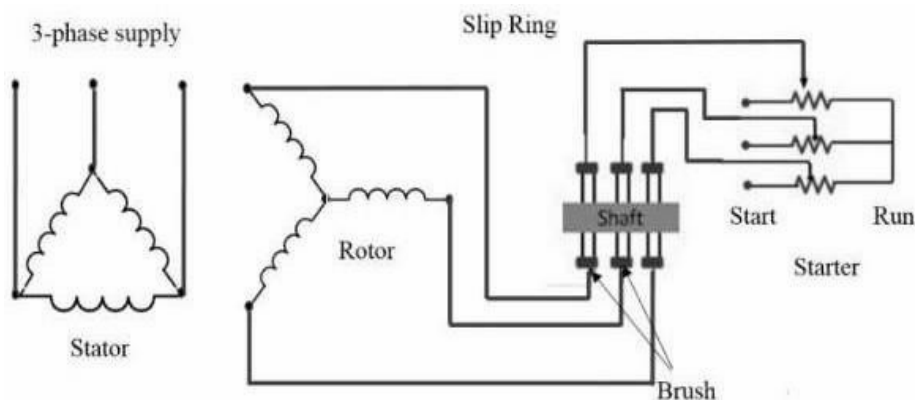
These slots assist to stop magnetic locking among the teeth of the stator & rotor so that smooth operation can be achieved and humming noise can be reduced. These motors include rotors of bars in place of the rotor winding where these bars are fabricated with brass, aluminum otherwise copper. In this type of motor, the winding of the rotor includes an aluminum bar otherwise uninsulated copper fixed into semi-closed rotor slots. At both ends of this motor, these conductors are short-circuited through similar material's end ring. As a result, this type of rotor is similar to a squirrel cage, so this is known as a squirrel cage induction motor.



SLIP RING INDUCTION MOTOR

The slip-ring induction motor is also called a wound rotor motor. In this motor, the rotor includes a laminated cylindrical core. Similar to the squirrel cage, there are some slots on the outer periphery. The rotor winding is placed inside the slots.

In the wound rotor, the insulated windings are wound on top of the rotor similarly like on the stator. The winding of this rotor can be distributed uniformly & usually connected in the STAR model. The three terminals of this STAR connection can be taken out throughout the slip ring. So this is the reason to call this motor a slip ring induction motor. Slip ring motors require external resistors to have high starting torque, low initial current and improved power factor. The rotor windings consist of more number of windings, higher induced voltage, and less current compared to the squirrel-cage rotor. The windings are connected to external resistance through slip rings, which helps to control the torque/speed of a motor. The rotor of this type of motor is wound type. It comprises of a cylindrical laminated steel core and a semi-closed groove at the outer boundary to accommodate a 3-phase insulated winding circuit.



Slip Ring Induction Motor Connection Diagram

The rotor is wound to match the number of poles on the stator. The three terminals of a rotor and three start terminals connecting through slip rings are connected to a shaft. The aim of the shaft is to transmit mechanical power.

Induction motors are used in industrial and domestic appliances because these are rugged in construction requiring hardly any maintenance, that they are comparatively cheap, and require supply only to the stator. The load is connected to the shaft.

Advantages and Disadvantages of Slip Ring Induction Motor

The advantages are

- High and excellent starting torque to support high inertia loads.
- It has a low starting current due to external resistance
- Can take full load current that is 6 to 7 times higher

The disadvantages are

- Includes higher maintenance costs due to brushes and slip rings compared to squirrel cage motor
- Intricate construction
- High copper loss
- Low efficiency and low power factor
- Expensive than 3 phase squirrel cage induction motor

Applications

Some of the **applications of slip ring induction motor** are

- These motors are used where higher torque and low starting current are required.
- Used in applications like elevators, compressors, cranes, conveyors, hoists

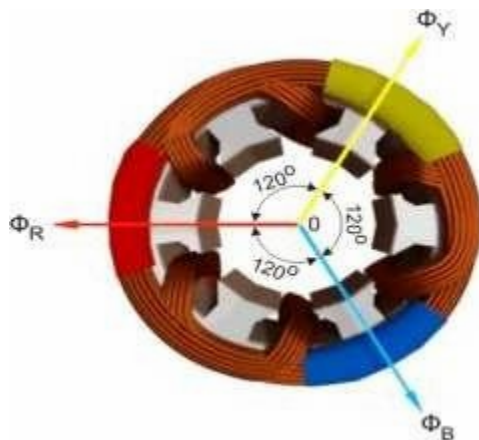
Comparison between Squirrel cage and Slip Ring Induction motor

Slip Ring Motor	Squirrel Cage Motor
It has a rotor of wound type	Its rotor is of squirrel cage type
Rotor has cylindrical core has parallel slots, in which each slot has a bar	Slots are not parallel to each other
Construction is complicated because of slip rings and brushes	Construction is simple
External resistance circuit is connected with a motor	No external resistance circuit as bars of the rotor is completely slotted
Starting torque is high	Torque is low
Efficiency is low	Efficiency is high

CONCEPT OF ROTATING MAGNETIC FIELD

When a three-phase supply is applied to a three-phase distributed winding of a rotating machine, a **rotating magnetic field** is produced which rotates in synchronous speed.

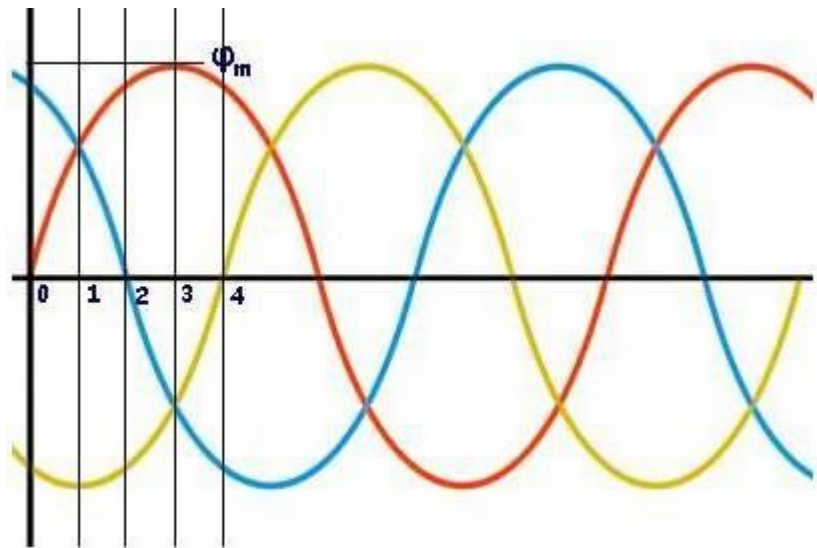
First consider one stator of an electric motor where three-phase winding is physically distributed in the stator core in such a manner that winding of each phase is separated from other by 120° in space.



Although the vector sum of three currents in a balanced three-phase system is zero at any instant, but the resultant of the magnetic fields produced by the currents is not zero rather it will have a constant non-zero value rotating in space in respect to time. The magnetic flux produced by the current in each phase can be represented by the equations given below. This is a similar representation of current is a three-phase system as the flux is cophasial with the current.

$$\begin{aligned}\phi_R &= \phi_m \sin(\omega t) \\ \phi_Y &= \phi_m \sin(\omega t - 120^\circ) \\ \phi_B &= \phi_m \sin(\omega t - 240^\circ)\end{aligned}$$

Where, ϕ_R , ϕ_Y and ϕ_B are the instantaneous flux of corresponding Red, Yellow and Blue phase winding, ϕ_m amplitude of the flux wave. The flux wave in the space can be represented as shown below.



Now, on the above graphical representation of flux waves, we will first consider the point 0.

Here, the value of ϕ_R is

$$\phi_R = \phi_m \sin(0) = 0$$

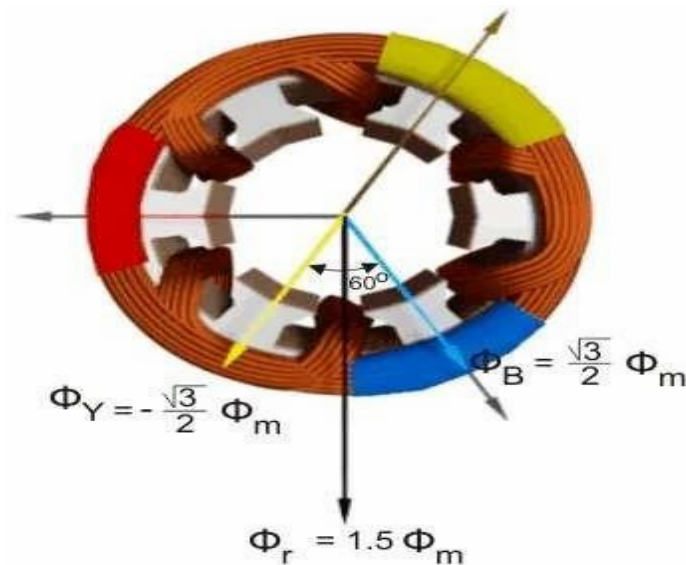
The value of ϕ_Y is

$$\phi_Y = \phi_m \sin(0 - 120^\circ) = \phi_m \sin(-120^\circ) = -\frac{\sqrt{3}}{2} \phi_m$$

The value of ϕ_B is

$$\phi_B = \phi_m \sin(0 - 240^\circ) = \phi_m \sin(-240^\circ) = \frac{\sqrt{3}}{2} \phi_m$$

The resultant of these fluxes at that instant (ϕ_r) is $1.5\phi_m$ which is shown in the figure below.



Now, on the above graphical representation of flux waves, we will consider the point 1, where $\omega t = \pi / 6$ or 30° .

Here, the value of ϕ_R is

$$\phi_R = \phi_m \sin(30^\circ) = \frac{1}{2} \phi_m$$

The value of ϕ_Y is

$$\phi_Y = \phi_m \sin(30^\circ - 120^\circ) = \phi_m \sin(-90^\circ) = -\phi_m$$

The value of ϕ_B is

$$\phi_B = \phi_m \sin(30^\circ - 240^\circ) = \phi_m \sin(-210^\circ) = \frac{1}{2} \phi_m$$

The resultant of these fluxes at that instant (ϕ_r) is $1.5\phi_m$ which is shown in the figure below. It is clear that the resultant flux vector is rotated 30° further clockwise without changing its value.

Now, on the graphical representation of flux waves, we will consider the point 2, where $\omega t = \pi/3$ or 60° .

Here, the value of ϕ_R is

$$\phi_R = \phi_m \sin(60^\circ) = \frac{\sqrt{3}}{2} \phi_m$$

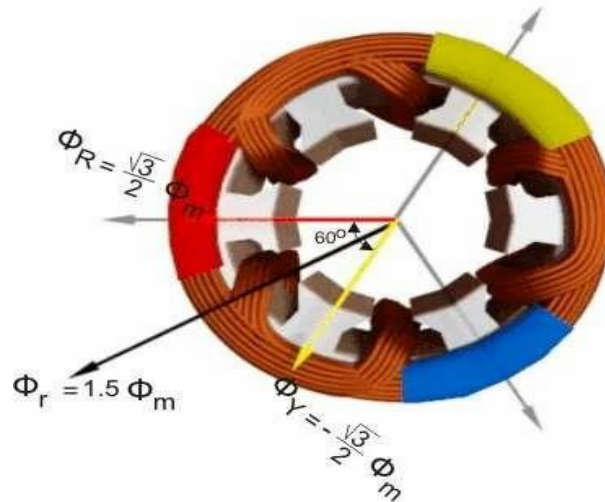
The value of ϕ_Y is

$$\phi_Y = \phi_m \sin(60^\circ - 120^\circ) = \phi_m \sin(-60^\circ) = -\frac{\sqrt{3}}{2} \phi_m$$

The value of ϕ_B is

$$\phi_B = \phi_m \sin(60^\circ - 240^\circ) = \phi_m \sin(-180^\circ) = 0$$

The resultant of these fluxes at that instant (ϕ_r) is $1.5\phi_m$ which is shown in the figure below. Here it is clear that the resultant flux vector is rotated 30° further clockwise without changing its value.



Now, on the graphical representation of flux waves, we will consider the point 3, where $\omega t = \pi / 2$ or 90° .

Here, the value of ϕ_R is

$$\phi_R = \phi_m \sin(90^\circ) = \phi_m$$

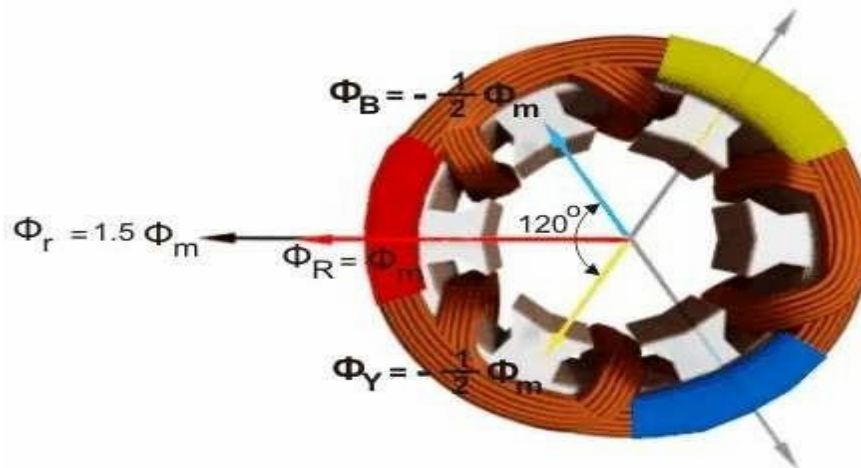
The value of ϕ_Y is

$$\phi_Y = \phi_m \sin(90^\circ - 120^\circ) = \phi_m \sin(-30^\circ) = -\frac{1}{2}\phi_m$$

The value of ϕ_B is

$$\phi_B = \phi_m \sin(90^\circ - 240^\circ) = \phi_m \sin(-150^\circ) = -\frac{1}{2}\phi_m$$

The resultant of these fluxes at that instant (ϕ_r) is $1.5\phi_m$ which is shown in the figure below. Here it is clear that the resultant flux vector is rotated 30° further clockwise without changing its value.



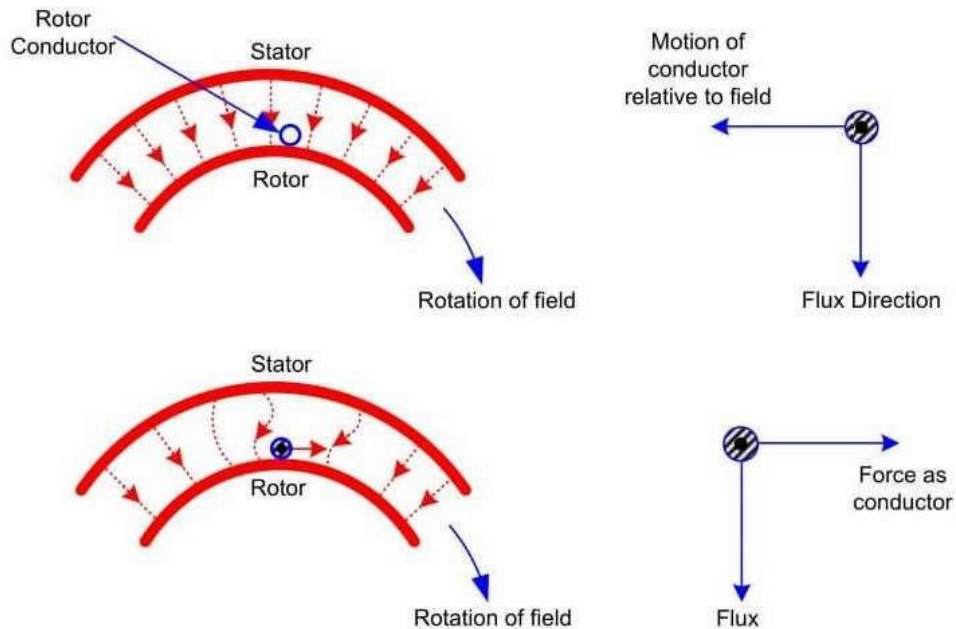
It can be proved that the due to balanced supply applied to the three phase stator winding a rotating or revolving magnetic fields is established in the space.

PRINCIPLE OF OPERATION OF THREE PHASE INDUCTION MOTOR:

The most commonly used AC motor which is also called asynchronous motor because this motor works at less speed as compared to synchronous speed. Here, synchronous speed is nothing but the rotary magnetic field's speed within the stator.

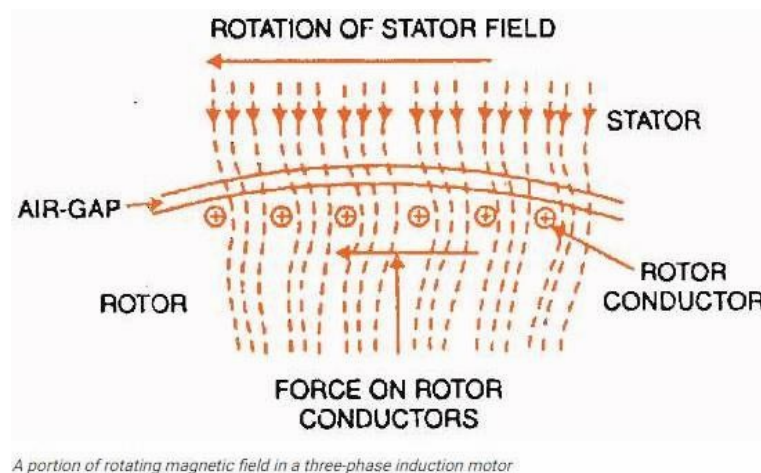
The induction motor works on the principle of induction where an electromagnetic field is induced into the rotor when the rotating magnetic field of the stator cuts the stationary rotor. Induction machines are by far the most common type of motor used in industrial, commercial, or residential settings.

Consider a portion of the three-phase induction motor as shown in the figure.



The working of the three-phase induction motor is based on the principle of electromagnetic induction.

When three-phase stator winding of an induction motor is energized from a 3 phase supply, a **rotating magnetic field** is set up which rotates around the stator at synchronous speed (N_s).



Synchronous Speed,

$$N_s = 120 f/P$$

Where,

f = frequency

P = Number of Poles

This rotating field passes through the air gap and cuts the rotor conductors, which are stationary. An EMF gets induced in every rotor conductor due to the relative speed between the rotating magnetic flux and the stationary rotor. Since the rotor circuit is short-circuited, currents start flowing in the rotor conductors.

The current-carrying rotor conductors are placed in the magnetic field produced by the stator. Consequently, a **mechanical force** acts on the rotor conductors. The sum of the mechanical forces on all the rotor conductors produces a **torque** which tends to move the rotor in the same direction as the rotating field.

The fact that the rotor is urged to follow the stator field (i.e., rotor moves in the direction of stator field) can be explained by Lenz's law.

According to Lenz's law, the direction of rotor currents will be such that they tend to oppose the cause of producing them. Now, the cause producing the rotor currents is the relative speed between the rotating field and the stationary rotor conductors. Hence to reduce this relative speed, the rotor starts running in the same direction as that of the stator field and tries to catch it. Hence three-phase induction motor starts running.

SLIP AND ITS SIGNIFICANCE IN INDUCTION MOTOR

In practice, the rotor can never reach the speed of stator flux. If it did, there would be no relative speed between the stator field and rotor conductors, no induced rotor currents and, therefore, no torque to drive the rotor.

The friction and windage would immediately cause the rotor to slow down. Hence, the rotor speed (N) is always less than the stator field speed (N_s). This difference in speed depends upon load on the motor.

The difference between the synchronous speed N_s of the rotating stator field and the actual rotor speed N is called **slip in a three-phase induction motor**.

Slip is usually expressed as a percentage of synchronous speed i.e.,

$$\text{Slip, } s = (N_s - N)/N_s \times 100 \%$$

The quantity $N_s - N$ is sometimes called **slip speed**.

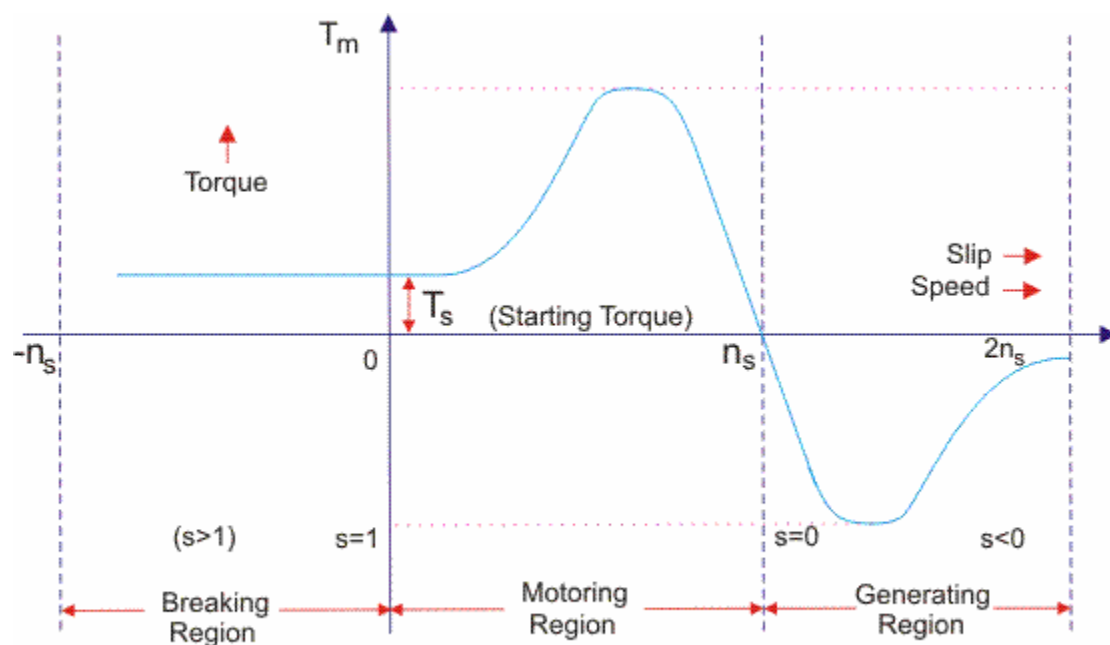
When the rotor is stationary (i.e., $N = 0$), slip, $s = 1$ or 100 %.

In an induction motor, the change in slip from no-load to full-load is hardly **0.1% to 3%** so that it is essentially a **constant-speed motor**.

TORQUE SLIP CHARACTERISTICS OF THREE PHASE INDUCTION MOTOR

The torque slip curve for an induction motor gives us the information about the variation of torque with the slip. The slip is defined as the ratio of difference of synchronous speed and actual rotor speed to the synchronous speed of the machine. The variation of slip can be obtained with the variation of speed that is when speed varies the slip will also vary and the torque corresponding to that speed will also vary.

The torque slip curve has three modes of operation as shown below.



Torque Slip Curve for Three Phase Induction Motor

Motoring Mode:

In this mode of operation, supply is given to the stator sides and the motor always rotates below the synchronous speed. The induction motor torque varies from zero to full load torque as the slip varies. The slip varies from zero to one. It is zero at no load and one at standstill. From the curve it is seen that the torque is directly proportional to the slip.

That is, more is the slip, more will be the torque produced and vice-versa. The linear relationship simplifies the calculation of motor parameter to great extent.

Generating Mode:

In this mode of operation induction motor runs above the synchronous speed and it should be driven by a prime mover. The stator winding is connected to a three phase supply in which it supplies electrical energy. Actually, in this case, the torque and slip both are negative so the motor receives mechanical energy and delivers electrical energy. Induction motor is not much used as generator because it requires reactive power for its operation.

That is, reactive power should be supplied from outside and if it runs below the synchronous speed by any means, it consumes electrical energy rather than giving it at the output. So, as far as possible, induction generators are generally avoided.

Braking Mode:

In the Braking mode, the two leads or the polarity of the supply voltage is changed so that the motor starts to rotate in the reverse direction and as a result the motor stops. This method of braking is known as plugging. This method is used when it is required to stop the motor within a very short period of time. The kinetic energy stored in the revolving load is dissipated as heat. Also, motor is still receiving power from the stator which is also dissipated as heat. So as a result of which motor develops enormous heat energy. For this stator is disconnected from the supply before motor enters the braking mode.

If load which the motor drives accelerates the motor in the same direction as the motor is rotating, the speed of the motor may increase more than synchronous speed. In this case, it acts as an induction generator which supplies electrical energy to the mains which tends to slow down the motor to its synchronous speed, in this case the motor stops. This type of breaking principle is called dynamic or regenerative breaking.

Problems on Induction Motors

Q1	A 3 phase, 4 pole, 440 V, 50 Hz induction motor runs with a slip of 4%. Find the rotor speed and frequency of the rotor current. $N_s = \frac{120 f}{P} = \frac{120 \times 50}{4} = 1,500 \text{ r.p.m.}$ $S = \frac{N_s - N}{N_s} \quad \text{i.e.} \quad 0.04 = \frac{1,500 - N}{1,500}$ $\therefore N = 1,440 \text{ r.p.m.}$ $f' = S f = 0.04 \times 50 = 2 \text{ Hz} = \text{frequency of rotor current}$
Q2	The frequency of the e.m.f. in the stator of a 4 pole induction motor is 50 Hz and that in the rotor is $1\frac{1}{2}$ HZ. What is the slip and at what speed the motor is running? $f' = S f \quad \text{i.e.} \quad \frac{3}{2} = S \times 50, \quad \therefore S = 0.03$ $N_s = \frac{120 f}{P} = \frac{120 \times 50}{4} = 1,500 \text{ r.p.m.}$ $S = \frac{N_s - N}{N_s} \quad \text{i.e.} \quad 0.03 = \frac{1,500 - N}{1,500}$ $\therefore N = 1455 \text{ r.p.m.}$
Q3	A 6 pole induction motor is connected to a 50 Hz supply. It is running at a speed of 970 r.p.m. Find the synchronous speed and the slip. $N_s = \frac{120 f}{P} = \frac{120 \times 50}{6} = 1,000 \text{ r.p.m.}$ $S = \frac{N_s - N}{N_s} = \frac{1,000 - 970}{1,000} = 0.03$

Q4

A 10 pole induction motor is supplied by a 6 pole alternator, which is driven at 1200 r.p.m. If the motor runs with a slip of 3%, what is its speed?

For alternator: $N = \frac{120 f}{P}$

i.e. $1200 = \frac{120 f}{6}$, $\therefore f = 60 \text{ Hz}$

For induction motor:

$$N_s = \frac{120 f}{P} = \frac{120 \times 60}{10} = 720 \text{ r.p.m.}$$

$$S = \frac{N_s - N}{N_s} \quad \text{i.e.} \quad 0.03 = \frac{720 - N}{720}$$

$$\therefore N = 698.4 \text{ r.p.m.}$$

Q5

A 4 pole, 50 Hz induction motor has a slip of 1 % at no load. When operated at full load, the slip is 2.5%. Find the change in speed from no load to full load.

On no load:

$$N_s = \frac{120 f}{P} = \frac{120 \times 50}{4} = 1,500 \text{ r.p.m.}$$

$$S = \frac{N_s - N}{N_s} \quad \text{i.e.} \quad 0.01 = \frac{1,500 - N_0}{1,500}$$

$$\therefore N_0 = \text{No load speed} = 1,485 \text{ r.p.m.}$$

On full load:

$$S = \frac{N_s - N}{N_s} \quad \text{i.e.} \quad 0.025 = \frac{1,500 - N}{1,500}$$

$$\therefore N = \text{Full load speed} = 1,462.5 \text{ r.p.m.}$$

$$\therefore \text{Change in speed} = N_0 - N = 1,485 - 1,462.5 = 22.5 \text{ r.p.m.}$$

Q6	A 12 pole, 3 phase alternator is coupled to an engine running at 500 r.p.m. It supplies an induction motor which has a full load speed of 1,440 r.p.m. Find the percentage slip and the number of poles of the motor. $f = \frac{PN}{120} = \frac{12 \times 500}{120} = 50 \text{ Hz (from alternator data)}$ N_s for induction motor is assumed to be 1,500 r.p.m. as the actual speed is 1,440 r.p.m. $N_s = \frac{120 f}{P} \quad \text{i.e.} \quad 1,500 = \frac{120 \times 50}{P}, \quad \therefore P = 4$ $\% \text{ slip} = \frac{N_s - N}{N_s} \times 100 = \frac{1,500 - 1,440}{1,500} \times 100 = 4\%$
Q7	A 12 pole, 3 phase, 50 Hz alternator is driven by a 440 V, 3 phase 6 pole induction motor running at a slip of 3%. Find the frequency of the e.m.f. generated by the alternator <i>For induction motor:</i> $N_s = \frac{120 f}{P} = \frac{120 \times 50}{6} = 1,000 \text{ r.p.m.}$ $N = N_s (1 - S) = 1000 (1 - 0.03) = 970 \text{ r.p.m.}$ The alternator is driven by induction motor. Hence, the speed of the alternator is 970 r.p.m. <i>For alternator :</i> $f = \frac{PN}{120} = \frac{12 \times 970}{120} = 97 \text{ Hz}$
Q8	A 3 phase, 4 pole, 400V, 50 Hz induction motor runs with a slip of 4%. Find the rotor speed and frequency. $N_s = \frac{120 f}{P} = \frac{120 \times 50}{4} = 1,500 \text{ r.p.m.}$ $S = \frac{N_s - N}{N_s} \quad \text{i.e.} \quad 0.04 = \frac{1,500 - N}{1,500} \quad \therefore N = 1,440 \text{ r.p.m.}$ $f' = S f = 0.04 \times 50 = 2 \text{ Hz}$

Q9	A 3 phase induction motor has 6 poles and runs at 960 r.p.m. on full load. It is supplied from an alternator having 4 poles and running at 1,500 r.p.m. Calculate the full load slip and the frequency of the rotor currents of the induction motor. $f = \frac{PN}{120} = \frac{4 \times 1500}{120} = 50 \text{ Hz (from alternator data)}$ For induction motor : $N_s = \frac{120 f}{P} = \frac{120 \times 50}{6} = 1,000 \text{ r.p.m.}$ $S = \frac{N_s - N}{N_s} = \frac{1,000 - 960}{1,000} = 0.04 \text{ or } 4\%$ $f' = S f = 0.04 \times 50 = 2 \text{ Hz}$
	Exercise Problems
	1 A 4 pole, 3 phase, 50 Hz induction motor is running on no load with a slip of 3%. Calculate the speed of the induction motor. (1,455 r.p.m.) 2 The rotor current of three phase, 4 pole, 50 Hz induction motor has a frequency of 2 Hz at full load. Calculate the slip and speed at full load. (0.04, 1,440 r.p.m.) 3 A 3 phase, 8 pole, 60 Hz induction motor has a rotor e.m.f. of 10 V induced in the rotor winding, at standstill. Calculate the voltage and frequency induced in the rotor winding when it rotates at 864 r.p.m. (0.4 V, 2.4 Hz) 4 An 8 pole induction motor is supplied by a 4 pole alternator running at 1,500 r.p.m. If the motor runs with a slip of 4%, what is its speed? (1,440 r.p.m.) 5 An 8 pole, 50 Hz induction motor has a slip of 2% on no load. On full load, the slip is 5%. Find the change in speed from no load to full load. (22.5 r.p.m.) 6 An 8 pole, three phase alternator is coupled to an engine running at 750 r.p.m. and supplies a three phase induction motor having a full load speed of 1,440 r.p.m. The frequency of the rotor current is 2 Hz. Find the percentage slip and the number of poles of the motor. (4%, 4) 7 A 4 pole, three phase induction motor runs at 1,470 r.p.m., when connected to a 50 Hz supply. Find the percentage slip and the frequency of the rotor currents. (2%, 1 Hz) 8 An 8 pole, three phase induction motor is supplied at 50 Hz and the frequency of the e.m.f. induced in the rotor is 2 Hz. Find the slip and speed of the motor. (0.04, 720 r.p.m.)

Alternators – Principle of Operation and Types

1. Introduction

- An **alternator** is an electromechanical device that converts **mechanical energy into electrical energy** in the form of **alternating current (AC)**.
- It is the **backbone of modern power generation** since almost all electricity produced worldwide is generated by alternators in power stations.
- Alternators are a special type of **AC generator**.
- They operate based on the principle of **electromagnetic induction** discovered by Michael Faraday in 1831.



2. Principle of Operation

- According to **Faraday's Law of Electromagnetic Induction**:

Whenever a conductor cuts the magnetic flux, an EMF (electromotive force) is induced in it, proportional to the rate of change of flux linkage.

Working in Alternators

1. The **rotor** carries a DC field winding. When excited by a DC supply, it produces a **magnetic field** (north and south poles).
2. The rotor is rotated by a **prime mover** (steam turbine, water turbine, diesel engine, or gas turbine).
3. As the rotor rotates, the **magnetic flux associated with stator conductors changes continuously**.
4. This change of flux linkage induces an **alternating EMF** in the **stator (armature) windings**.
5. The induced EMF is **sinusoidal** due to the uniform rotation of the magnetic field.

Frequency of Induced EMF

$$f = \frac{P \times N}{120}$$

Where:

- f = frequency of induced EMF (Hz)
- P = number of poles on the rotor
- N = speed of rotor in rpm

👉 Example:

If a 4-pole alternator runs at 1500 rpm, frequency = $(4 \times 1500)/120 = 50 \text{ Hz}$.

3. Construction of Alternator

Alternators are made up of two main parts: **stator (stationary)** and **rotor (rotating)**.

A. Stator (Stationary Part)

- Consists of:
 - **Stator frame** → provides mechanical support.
 - **Stator core** → laminated silicon steel to reduce eddy current losses.
 - **Slots** → cut on inner surface to house armature windings.
 - **Armature winding** → 3-phase windings (star or delta connected) where EMF is induced.

B. Rotor (Rotating Part)

- Carries **field winding** excited by DC.
- Types of rotors:
 1. **Salient Pole Rotor** → Used in hydro power stations.
 2. **Non-Salient (Cylindrical) Rotor** → Used in thermal and nuclear stations.

C. Auxiliary Parts

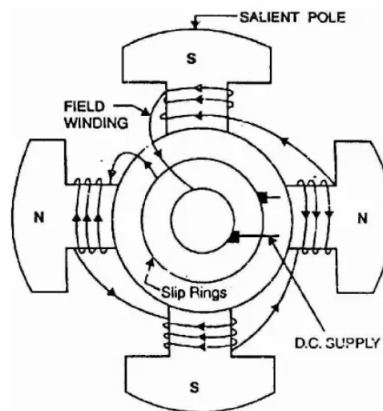
- **Bearings** → for smooth rotation.
- **Exciter** → provides DC supply to rotor.
- **Cooling system** → air, hydrogen, or water cooling to dissipate heat.
- **Slip rings & brushes** → used in DC excited alternators to feed DC into rotor winding.

4. Types of Alternators

A. Based on Rotor Construction

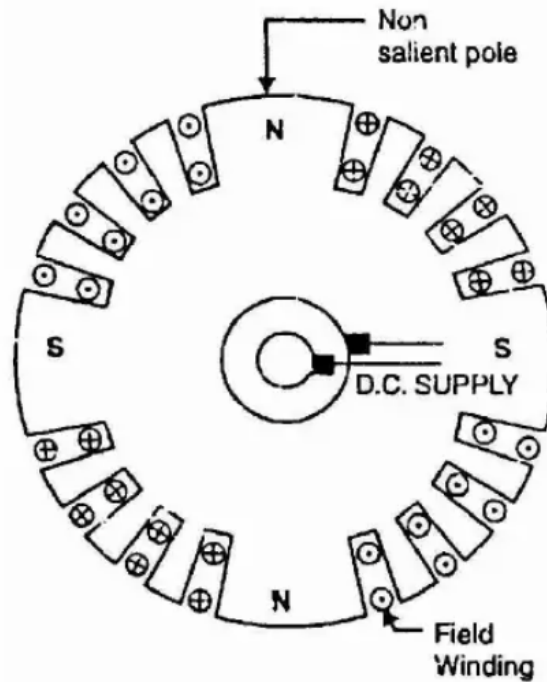
1. Salient Pole Alternators

- Poles project out (salient = projecting).
- Large diameter, short axial length.
- Low/medium speed machines (100–400 rpm).
- Mostly used in **hydro power stations**.
- Advantages: High efficiency at low speed, easy cooling.
- Disadvantages: Cannot run at very high speeds due to mechanical stress.



2. Non-Salient (Cylindrical Pole) Alternators

- Rotor is smooth and cylindrical with slots for field winding.
- Small diameter, long axial length.
- High-speed machines (3000 rpm).
- Used in **thermal and nuclear stations**.
- Advantages: Suitable for high speed, mechanically robust, low windage loss.



5. Advantages of Alternators

- High efficiency for large-scale power generation.
- Can generate electricity at high voltages (11–33 kV) and frequencies (50/60 Hz).
- Robust design with long life.
- Suitable for continuous operation in power plants.

6. Applications of Alternators

- Power generation in hydro, thermal, nuclear, and diesel power stations.
- Automotive alternators for charging vehicle batteries.
- Emergency generators for hospitals, industries, data centers.
- Used in renewable energy systems (wind turbines, small hydro).